



Mapping coastal green infrastructure along the Pondicherry coast using remote sensing data and machine learning algorithm

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Coastal green infrastructure provides numerous ecosystem services, including flood protection, erosion control, carbon sequestration, and habitat for marine life. Mapping and monitoring these critical coastal habitats are essential for sustainable management and conservation efforts. This study employed the Google Earth Engine (GEE) cloud platform with high-resolution multispectral satellite imagery (Sentinel-2 data with 10 m spatial resolution), Airborne Laser Terrain Mapper (ALTM) elevation data with 5 m resolution, and advanced machine learning (ML) algorithms used to map the distribution and extent of coastal green infrastructure along the Pondicherry coastline in southern India. A random forest (RF) classifier was trained on a diverse set of reference data (70% for training and 30% for validation) collected through extensive field surveys and visual interpretation of very high-resolution aerial imagery. The model integrated spectral information from multiple satellite sensors along with derived biophysical indices to accurately delineate different coastal vegetation types. The resulting maps revealed detailed spatial patterns of mangroves, sand dunes, and coastal plantations with an overall accuracy exceeding 90% verified with the field data. Analyses quantified their spatial coverage and fragmentation along the study area. This high-resolution, accurate baseline data can inform coastal management strategies, including targeted conservation efforts, ecological restoration projects, climate change adaptation planning, and sustainable development practices that preserve vital green infrastructure. The workflow demonstrated the robust capabilities of ML methods coupled with multi-source remote sensing data for effectively mapping complex and dynamic coastal ecosystems at a regional scale. The techniques can be adapted for other coastal regions to understand green infrastructure dynamics better and support evidence-based policies promoting ecological and community resilience.

Keywords. Mangrove; sand dune; coastal plantation; Remote Sensing; ALTM.

1. Introduction

Coastal green infrastructure (CGI), such as mangroves, salt marshes, and seagrass meadows, plays a crucial role in providing ecosystem services and supporting coastal resilience (Barbier *et al.* 2011;

Spalding *et al.* 2014). These ecosystems act as natural buffers against extreme weather events, mitigating the impacts of storms, floods, and erosion (Gedan *et al.* 2011). Additionally, CGI contributes significantly to carbon sequestration, helping to mitigate climate change (McLeod *et al.*

2011). Furthermore, these ecosystems serve as nursery grounds for various marine species and support biodiversity, contributing to the overall health and productivity of coastal and marine environments (Beck *et al.* 2001; Nagelkerken *et al.* 2008).

Despite their significant ecological and socioeconomic importance, coastal green infrastructure is facing unprecedented threats from human activities and climate change (Duarte *et al.* 2008). Urbanization, coastal development, pollution, and unsustainable resource exploitation have led to the degradation and loss of these valuable ecosystems (Valiela *et al.* 2001). Climate change-induced factors, such as sea-level rise, increased storm intensity, and ocean acidification, further exacerbate the pressures on CGI (Gilman *et al.* 2008). Consequently, there is an urgent need for effective monitoring, mapping, and management strategies to conserve and restore these vital ecosystems.

Remote sensing technologies, coupled with machine learning algorithms, have emerged as powerful tools for mapping and monitoring CGI over large spatial scales (Heumann 2011; Kuenzer *et al.* 2011). Satellite imagery, with its ability to capture data across various spatial and temporal resolutions, offers a comprehensive and cost-effective approach to assessing the distribution, extent, and health of CGI (Roelfsema *et al.* 2018). Machine learning algorithms, such as random forests, support vector machines, and deep learning models, have demonstrated remarkable performance in classifying and mapping coastal ecosystems from remote sensing data (Traganos and Reinartz 2018; Pham *et al.* 2019).

The Pondicherry coast, located along the Coromandel Coast of the Bay of Bengal, is a region of significant ecological and economic importance. This coastal stretch is home to diverse CGI, including mangrove forests, seagrass meadows, and salt marshes, which support various ecosystem services and livelihoods (Kathiresan 2012). However, these ecosystems face threats from urbanization, pollution, and climate change-induced sea-level rise (Gnanappazham *et al.* 2017), highlighting the need for effective monitoring and management strategies.

In this study, we aim to leverage the power of remote sensing data and machine learning techniques to map and assess the spatial distribution of CGI along the Pondicherry coast. By combining high-resolution satellite imagery with advanced classification algorithms, we aim to provide an

accurate and up-to-date inventory of CGI in the region. This information can inform coastal management decisions and conservation efforts and contribute to the sustainable development of the region.

Specifically, we will employ multispectral satellite data to capture the unique spectral and structural characteristics of different CGI types (Goudenhoofd and Delobbe 2009; Pham and Yunus 2020). Machine learning algorithms, such as random forests and deep learning models, will be trained on a subset of the data to learn the patterns and features associated with each CGI class (Traganos *et al.* 2018; Kou *et al.* 2020). The trained models will then be applied to the entire study area to generate detailed maps of CGI distribution.

Our study aims to address the following objectives:

1. Accurate mapping and classification of coastal green infrastructure, including mangroves, sand dunes, and coastal plantations.
2. Assessment of the extent of these ecosystems.
3. Identification of areas vulnerable to degradation or loss.
4. The study contributes to the development of conservation and management strategies for the Pondicherry coastal green infrastructure.

The resulting maps and spatial information will provide valuable insights into the current state of CGI along the Pondicherry coast, enabling decision-makers and stakeholders to identify priority areas for conservation and restoration efforts.

2. Materials and method

2.1 Study area

Pondicherry coastal region, situated on the Coromandel Coast of the Bay of Bengal, is located between latitudes 11°46' and 12°30' towards the north and longitudes 79°36' and 79°53' towards the east. It encompasses the Union Territory of Pondicherry and extends from the northern village of Ganagachettikulam to the southern village of Nallavadu. It is a multifaceted zone that encompasses rich biodiversity, cultural diversity, and historical heritage. The coastal stretch is approximately 24 km long and is endowed with a diverse range of ecosystems, including beaches, mangrove forests, salt marshes, and lagoons. Understanding and preserving the unique attributes of this coastal

region is imperative for sustainable development and conservation efforts. The present study has implemented a 500 m area along the Pondicherry coast from the shoreline. The average elevation of the study site is about 15 m from the mean sea level (Management 2020). Pondicherry needs this study to safeguard the coastline, and study findings offer protection against erosion and floods, which can be brought on by winds, currents, and tides. The map of the study area is shown in figure 1.

2.2 Remote sensing data

2.2.1 Sentinel-2

Our study utilized Sentinel-2A&B data as detailed in table 1, which provides ortho-rectified and atmospherically corrected information in multiple spectral bands. The data were acquired through the Copernicus Open Access Hub and were available free, allowing us to conduct cost-effective environmental monitoring projects. We selected specific acquisition dates and areas of interest relevant to our study. Prior to analysis, we performed essential pre-processing steps on the Sentinel-2 data. These included radiometric calibration, cloud masking, and geometric correction. These steps were crucial to ensure that our analysis was not influenced by atmospheric disturbances or cloud cover, which can significantly influence the quality of the data.

2.2.2 Normalized Differential Vegetation Index (NDVI)

NDVI is a commonly used remote sensing technique to assess and monitor the health and vitality of vegetation. NDVI is calculated using the following formula:

$$\text{NDVI} = (\text{NIR} - \text{Red}) / (\text{NIR} + \text{Red})$$

where NIR is the Near InfraRed band of the electromagnetic spectrum. Red is the Red band of the electromagnetic spectrum.

By taking the difference between the NIR and Red reflectance values and dividing it by their sum, NDVI values can be generated that range from -1 to 1.

2.2.3 Airborne Laser Terrain Mapper (ALTM)

ALTM technology has revolutionized the way we collect and analyse geographical data. ALTM data,

derived from Light Detection and Ranging (LiDAR) sensors mounted on aircraft, provide high-resolution, three-dimensional representations of the Earth's surface. This journal write-up explores the key aspects of ALTM data, including its acquisition, processing, and various applications. ALTM data acquisition involves the deployment of LiDAR sensors on an aircraft, which emits laser pulses towards the Earth's surface and records the time it takes for these pulses to return. The data obtained in this process can be divided into two main categories: point clouds and intensity data. Point clouds consist of millions of individual data points, each representing a specific location on the Earth's surface. Intensity data provides information about the reflectivity of the surface at each point. Processing ALTM data is a complex but crucial step in extracting meaningful information. Raw point cloud data typically require cleaning to remove outliers and errors caused by factors like atmospheric interference. Algorithms are then applied to classify the points into different categories, such as ground, vegetation, and buildings. These classified point clouds can be further processed to create Digital Elevation Models (DEMs) and Digital Surface Models (DSMs), which are valuable for a wide range of applications, including topographic mapping, flood modelling, and urban planning. For a few specific locations along the Indian coastline, ALTM data have been gathered at 5 m resolution and up to two kilometers from the coast (Nayak *et al.* 2012). These data were used in the current investigation along the Pondicherry coast.

2.3 Machine learning

Satellite imagery (e.g., Landsat, Sentinel-2) and ancillary data (e.g., elevation, climate) were retrieved from the GEE platform. Performed necessary pre-processing steps such as atmospheric correction, image calibration, and cloud masking to prepare the data for analysis. Collected a representative set of training samples for different land cover classes, including plantations, using ground truth data and existing land cover maps. Extracted relevant spectral, textural, and contextual features from satellite imagery, which serve as input variables for the ML algorithm (RF) shown in figure 2. Trained supervised ML algorithms using the labelled training data to build a predictive model capable of classifying plantation areas accurately.

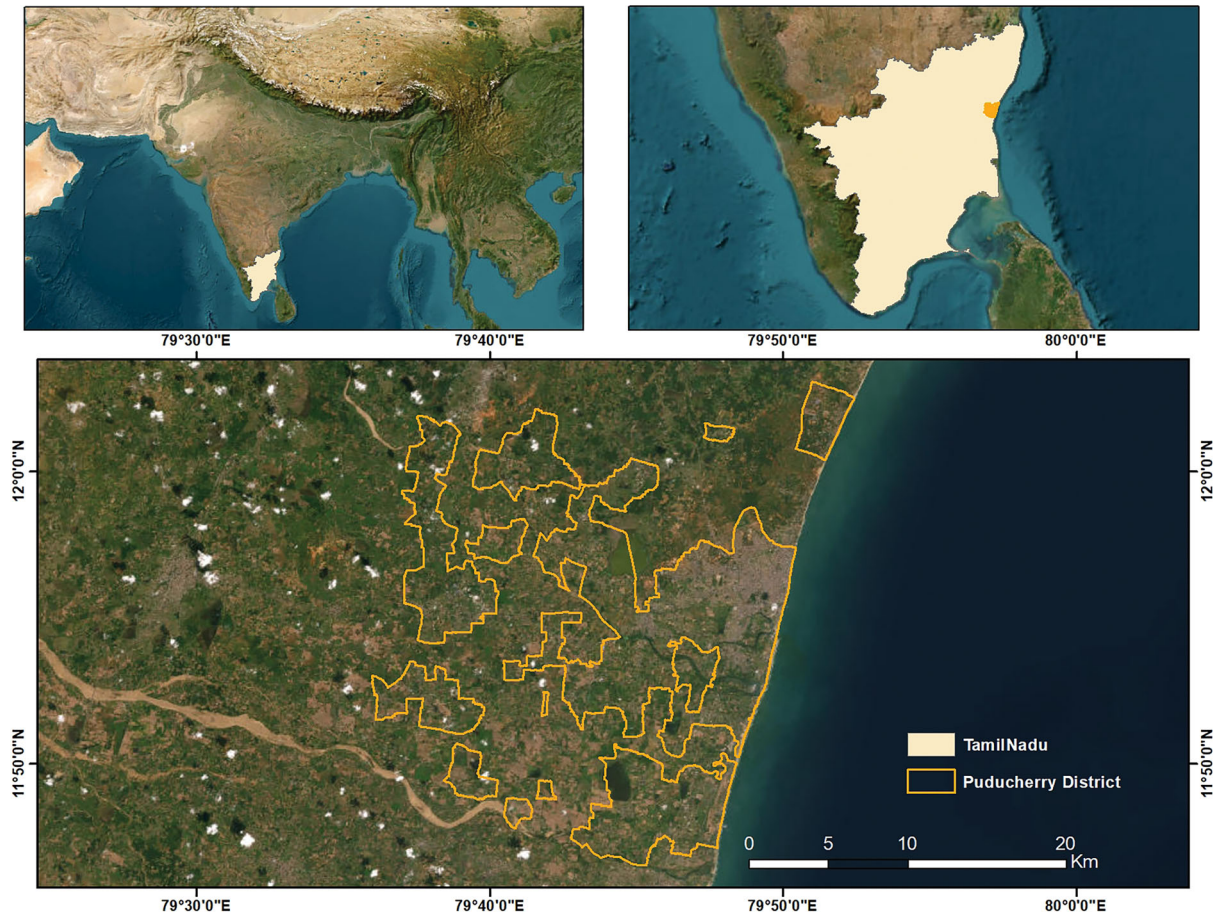


Figure 1. Study area location map.

Applied the trained model to classify the entire study area into different land cover classes, including plantations. Conduct post-classification processing to refine the results, such as spatial filtering, majority filtering, and accuracy assessment.

2.3.1 Random forest

Random forest has been widely used for satellite data classification due to its ability to handle high-dimensional data and its robustness to noise and outliers. Numerous studies have demonstrated the effectiveness of random forest for satellite data classification tasks, often outperforming traditional classifiers such as maximum likelihood or support vector machines (Gislason *et al.* 2006; Rodriguez-Galiano *et al.* 2012; Belgiu and Drăguț 2016). Random forest is an ensemble learning method that combines multiple decision trees to improve predictive accuracy and reduce overfitting (Breiman 2001). It has been

successfully applied to classify satellite data, such as land cover/land use (LULC) mapping, crop type identification, and urban area extraction (Belgiu and Drăguț 2016).

In the context of satellite data classification, the random forest algorithm operates as follows:

1. From the input satellite imagery and associated ground truth data, a bootstrap sample is created by randomly sampling pixels with replacement.
2. For each bootstrap sample, an unpruned decision tree is grown using a random subset of the spectral bands or features at each node.
3. Steps 1 and 2 are repeated to grow a large number of decision trees (e.g., hundreds or thousands).
4. For a new pixel, the final classification is obtained by aggregating the predictions from all the individual trees in the ensemble, typically using majority voting for the most frequent class.

Table 1. Sentinel-2A&B data band details.

Name of the sensor	Platform	Spectral bands	Central wavelength (nm)	Bandwidth (nm)	Spatial resolution (m)	Data source
Sentinel-2A&B	Satellite	Band 1 – Coastal aerosol	442.2	21	60	Google Earth Engine
		Band 2 – Blue	492.1	66	10	
		Band 3 – Green	559.0	36	10	
		Band 4 – Red	664.9	31	10	
		Band 5 – Vegetation red edge	703.8	16	20	
		Band 6 – Vegetation red edge	739.1	15	20	
		Band 7 – Vegetation red edge	779.7	20	20	
		Band 8 – Near InfraRed (NIR)	832.9	106	10	
		Band 8A – Narrow NIR	864.0	22	20	
		Band 9 – Water vapour	943.2	21	60	
		Band 10 – SWIR – Cirrus	1376.9	30	60	
		Band 11 – SWIR	1610.4	94	20	
		Band 12 – SWIR	2185.7	185	20	

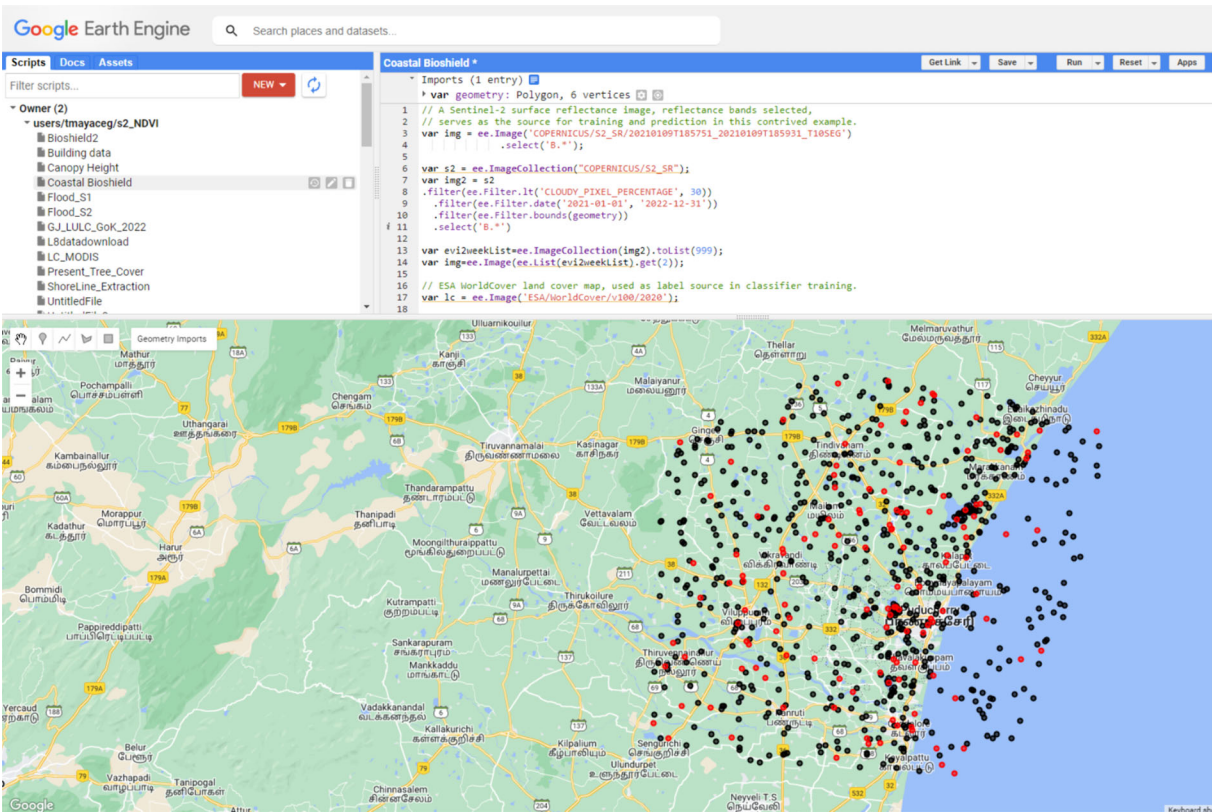


Figure 2. GEE Platform to run machine learning for plantation mapping (red dots denote training points, and black dots denote validation points).

2.4 Field data

The comprehensive field survey was conducted along the 24 km stretch of Pondicherry coast. The

extended landward area of Thengaithittu estuary and backwater (Pondy-necklace) was visited to record the status of mangroves along the backwaters. The pre-identified sand dune locations

through satellite data extraction were verified through ground truth. The hand-held Garmin GPS was used to record the location of the dune during the survey. The length, width, and height of the dune were measured by using the Laser range finder (Nikon Forestry Pro). The visual field survey was conducted by walking along the shore to collect the major plant species on beach and coastal stretches. The status of coastal protection sand dunes, plantation types (natural/manmade), vegetation area cover, maximum sand dune height, natural and anthropogenic impacts were noted through encrypted images with geo-locations. Figure 3 demonstrates the methodology flow chart for the study.

3. Results and discussion

3.1 Sand dune mapping

The coastal sand dune ecosystem is one of the major components that play a significant role in protecting the interface between land and sea. Sand dunes were extracted from the high-resolution ALTM data. The first and foremost step is to determine a suitable elevation threshold for sand dunes. Sand dunes will typically have elevations higher than the surrounding areas. Under the purview of coastal protection, sand dunes, especially parallel dunes, act as first-level coastal protection structures against natural hazards. In India, along the coast of Tamil Nadu, 5-foot (~ 1.5 m) tall sand dunes are seen to provide protection to habitations 500 m away from the sea (Namboothri *et al.* 2008). In Welsh landscapes, Pye *et al.* (2017) provide an elaborate diagrammatic depiction of low coastal protection provided by sand dunes <2 m high. Considering factors such as seasonal variations in dune height and erosion, the least possible threshold for sand dunes to act as coastal protectors at a distance of up to 500 m from the shoreline was chosen as >1 m. Moreover, the contour intervals considered for this study is 1 m, thereby resulting in a minimum dune height of 2 m. Several strategies to create sand dunes as coastal protectors using coastal fencing also consider a height of 1.5 m or more, thereby supporting our choice of minimum required height for sand dunes to serve the said purpose (Wootton *et al.* 2016; Eichmanns *et al.* 2021). The present study considers more than 1 m elevation extent as a dune. The training data was collected from ground (field) and non-ground (Google Earth Platform). A

clipping tool was used to filter out the buildings and agricultural lands from the ALTM data to distinguish the sand dunes. This step helps to remove vegetation and buildings from the data. Results have been validated by comparing them with ground truth data and assessing the accuracy of the extracted dune features. Ground truth photos are shown in figure 4. The analysis revealed the spatial distribution and characteristics of sand dunes along the Pondicherry coast, shown in figure 5 and the statistics of sand dunes on the Pondicherry coast, shown in table 2. Dune heights, orientations, and morphological parameters were quantified, providing valuable insights into dune dynamics and evolution.

Additionally, the comparison with ground truth data demonstrated the reliability and accuracy of the ALTM-derived sand dune maps. The back-shore, fore-dune and frontal dune slope of Pondicherry was widely covered by *Ipomoea* sp. and *Spinifex* sp., followed by herbs and shrubs such as *Euphorbia peplis*, *Bulbostylis barbata*, *Euphorbia granulate*, *Mollugo verticillata*, *Ocimum americanum*, *Polygonum aviculare* and grass species are found in the sandy shore and dune environment (figure 6). Sand dunes were observed in the southern part of the Pondicherry coast. The natural dunes were 2 to 5 m in height, and the width varies from 8 to 47 m along the coast from Mullodai lagoon mouth near Thengaithittu estuary mouth. However, the Kalapet town of Pondicherry coast has a wider beach, and a minimal area of coastal sand dunes were observed, and Manapet village of Pondicherry coast has a more sand dune area with height varying from 2 to 10 m. The manmade sand dunes of 6–8 m in height, 64–140 m in width and length of 208.7–327 m parallel to the coastline were observed in the Thengaithittu harbour mouth. These manmade sand dunes are the dredged soil dumps. Later, established plants *ipomoea*, grass species, and *Prosopis* sp. naturally dominated the dune and stabilized the sand dune structure (figure 4a).

3.2 Mangrove mapping

This study used pixel-based classification and vegetation indices to map the mangrove (Jia *et al.* 2021). The classification identified various distinct land cover classes (water, built-up, vegetation, etc.), including various mangrove species like

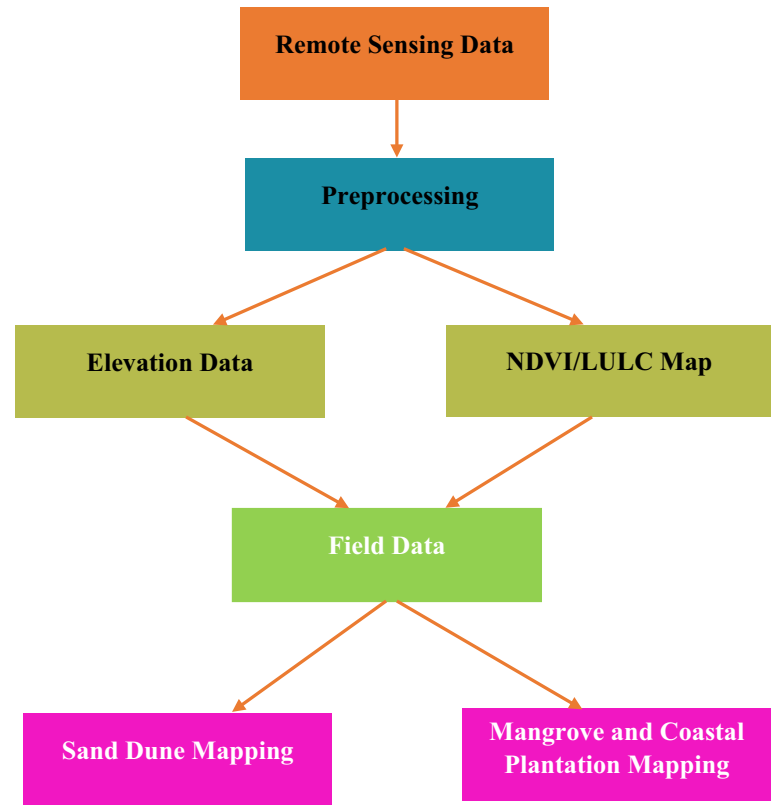


Figure 3. Methodology flow chart for the present study.



Figure 4. (a) Established natural vegetation on manmade sand dunes. (b) Anthropogenic human pedestrian traffic. (c) Unorganized boat and vehicle parking impact.

Avicennia, *Rhizophora*, *Bruguiera* and associated non-mangrove features. Figure 7 shows the field photos of different mangrove species in the study area. Mullodai lagoon and Uppanar estuary and backwater area were covered by moderate to dense mangrove and mangrove-associated plants. Panithittu and Nallavadu backwater mangroves are less dense as well, and *Prosopis* species domination was observed. Pooranankuppam was located in the middle of the Chunnambar River.

Thengaithittu estuary and the adjacent backwater area were densely covered with mangrove species. The total mangrove cover in Pondicherry was estimated at 48 ha along the Thengaithittu, and the spatial map is shown in figure 8. Further analysis revealed spatial distribution patterns of different mangrove species, highlighting areas with dense patches and fragmented ones. The classified map was verified with the field observation data.

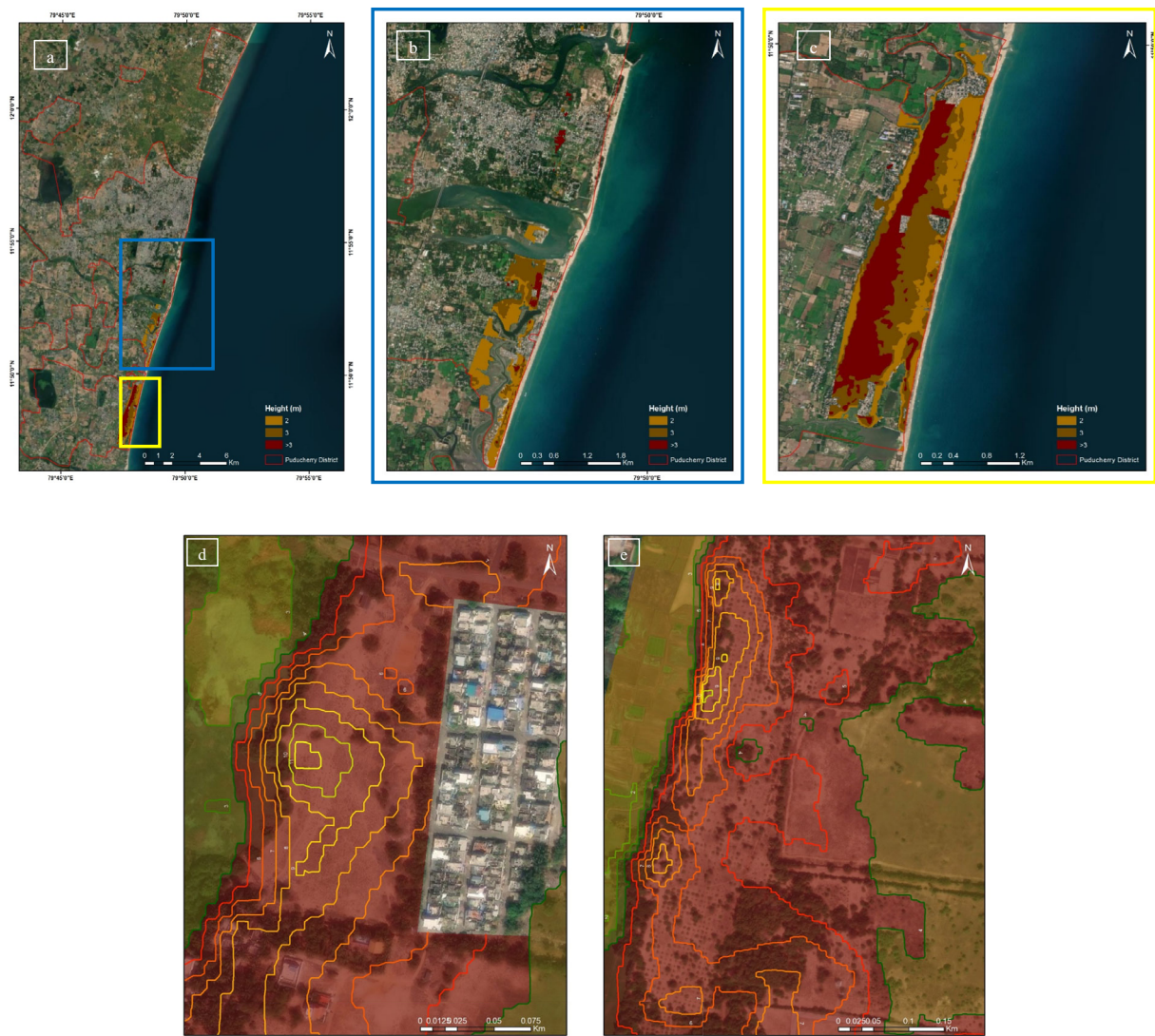


Figure 5. (a, b, c) Sand dune map for the study area; (d, e) Micro-level sand dune mapping.

Table 2. Village-wise statistics for sand dunes of Pondicherry.

Name of the village	Min height (m)	Max height (m)	Area (ha)
Ariankuppam	2	5	5.6
Kalapet	3	4	0.3
Kirumampakkam	2	10	63.2
Manapet	2	10	279.3
Manaveli	2	4	0.8
Pillaichavady	2	4	0.6
Pillayarkuppam	3	11	4.4
Pooranankuppam	2	8	139.7
Puducherry	2	3	0.5
Thengaithittu	2	8	5.8
Total area in ha			500.2

3.3 Coastal plantation mapping

The application of ML techniques on GEE facilitated the accurate mapping of coastal plantations,

providing valuable insights into their distribution and characteristics. The ML model output was validated with the field data and field photographs shown in figure 9. The generated map can support

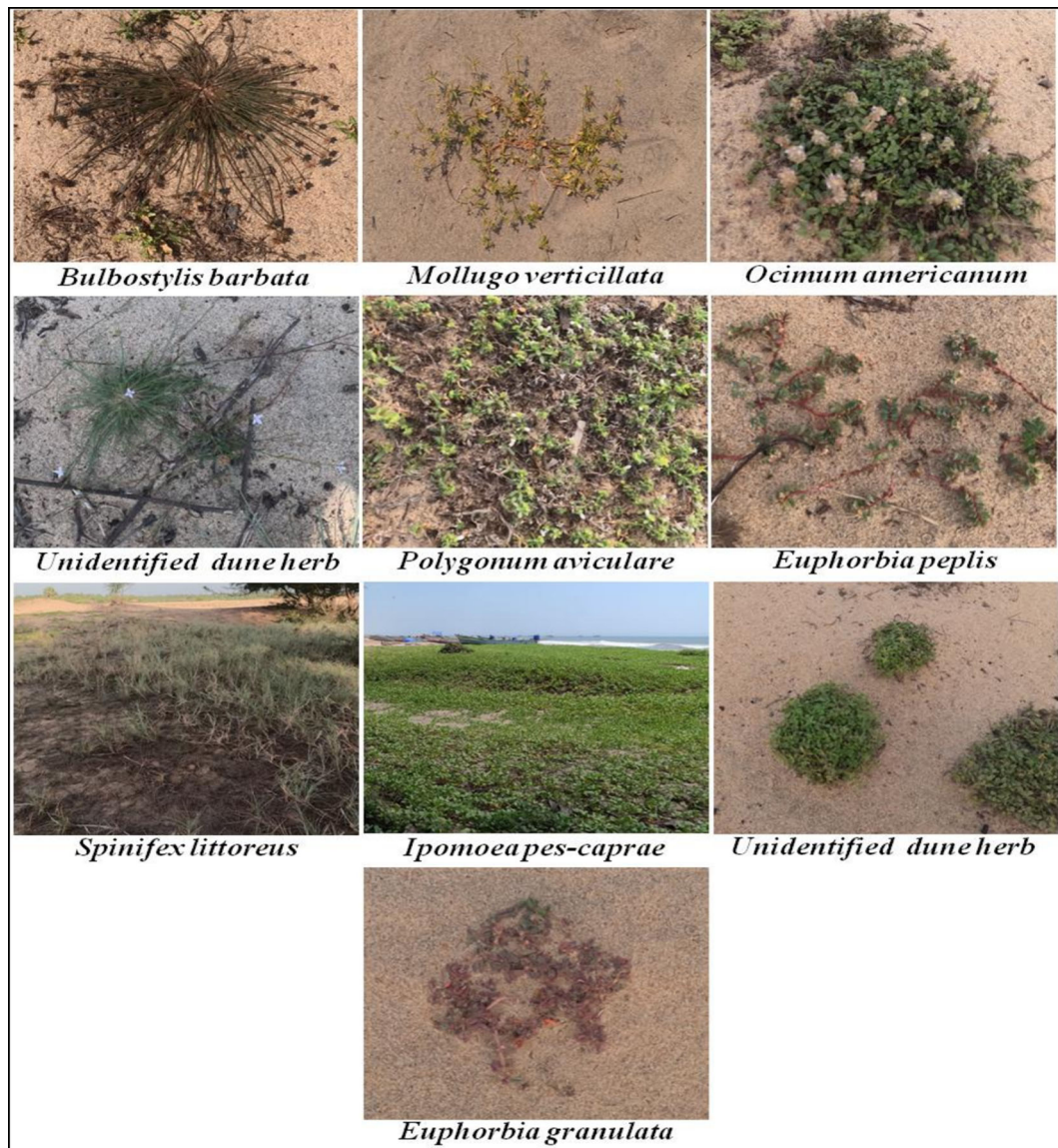


Figure 6. Dominant herbs and shrubs species found on the sandy shore and coastal sand dunes of the Pondicherry district.



Figure 7. Distribution of mangrove species on the sides of Pandy-necklace water channel (1. *Avicennia* sp., 2. *Rhizophora* sp., 3. *Bruguiera* sp.).

decision-making processes related to coastal land management, conservation efforts, and resource allocation. The *Cocos nucifera* (Coconut tree) was the predominant tree species that occupied ~ 36.8 ha of coastal land, followed by *casuarina* plantation ~ 26.1 ha, *Borassus flabellifer* (Palm tree) ~ 8.2 ha and *Anacardium occidentale* (Cashew tree) 3.5 ha. These mono-tree types are the dominant species found along the foreshore and adjacent near shore that contribute to coastal protection (figure 9). The notable invasive *Prosopis* occupied ~ 19.07 ha of Pondicherry coastal land. Approximately 125.1 ha of coastal plant cover was found with mixed vegetation (table 3). A major portion of this mixed vegetation is covered by *Thespesia*

Populnea (Teak tree) and *Acacia auriculiformis* (Eucalyptus) plantations on private lands with other small trees, shrub trees, and larger trees such as *Azadirachta indica* (Neem), and *Millettia pinata* (Pongamia) species on the land boundaries (figure 10). Also, the Neem and Pongaima scatterly grew in the urban areas adjacent to the coast. From Ariankuppam to Thengaithittu, the back-shore hinterland is protected by vegetation cover, and the sandy beach was observed. The southern part of Pondicherry and Kalapet village coast observed a wider sandy beach (about 16–47 m) followed by a moderate density of coastal green belt vegetation. From Pudhukuppam Beach's northern side to the southern side of Paradise



Figure 8. Mangrove mapping for the study area.

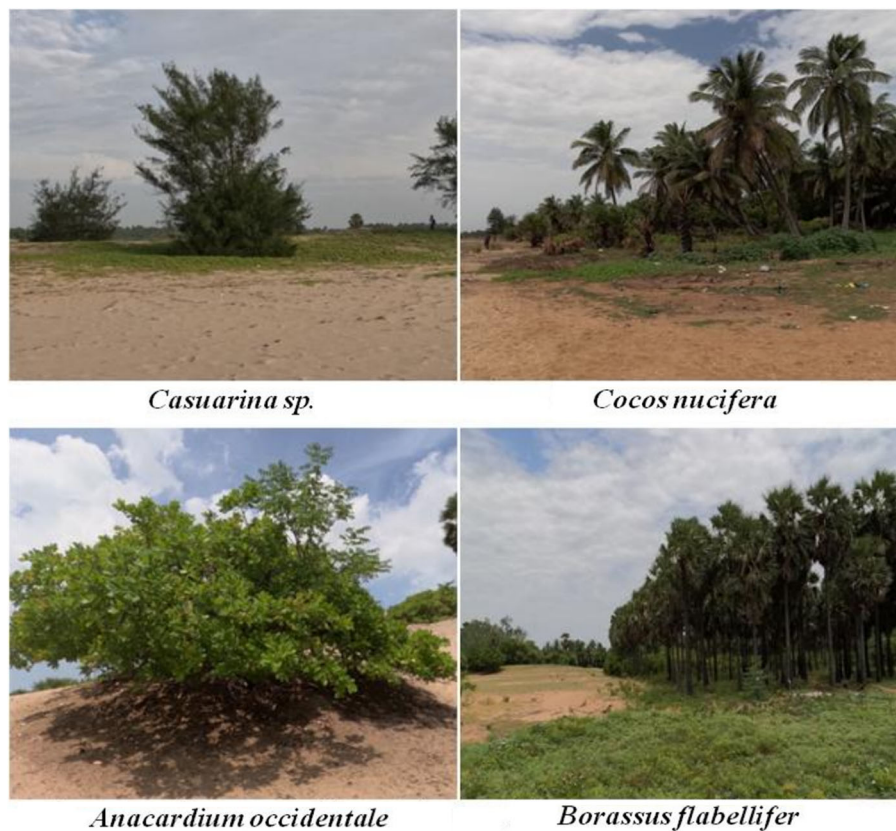


Figure 9. Dominant coastal protection tree species found on the sandy shore and coastal sand dunes of the Pondicherry district.

Beach, the Tengaithittu plantation (*Borassus flabellifer* and *Casuarina sp.*) was observed in less to moderate density in the field. In the continuous pattern of *Cocos nucifera*, *Anacardium occidentale* species of the manmade plantation was observed along the sandy coastal stretch. The Eden Beach to Veerampattinam Beach, in the southern part of the shore, was dominated by *Casuarina sp.* plantation. The southern part of Pondicherry and Kalapet village coast are observed to have wider sandy beaches (about 16–47 m), followed by a moderate density of coastal green belt vegetation. The Uppanar harbour mouth to Sodhanaikuppam village Kottakuppam Beach was protected by an erosion protection Rubble Mound Sea (RMS) wall. In Pondicherry, the urban coastal stretch observed an absence of shelterbelt vegetation. The water-logged areas, such as ponds and near shore waters, were dominated by *Fimbristylis annua* and *Eichhornia crassipes* species. The plantation map along the coastal area is shown in figure 11.

3.4 Accuracy assessment

Accuracy is a fundamental aspect of any measurement or evaluation process, as it directly impacts

the validity and reliability of the results obtained. In scientific research, experimental studies, and data analysis, ensuring high levels of accuracy is critical for drawing meaningful conclusions and making informed decisions. Accuracy refers to the closeness of a measured or calculated value to its true or accepted value (Taylor and Thompson 1982). It is a measure of how well an instrument, method, or system is able to produce results that are consistent with the actual or reference value. Accuracy is often expressed as a percentage or absolute value, indicating the degree of deviation from the true or accepted value (Natrella 2010). However, high precision does not necessarily imply high accuracy, as the results may be consistently biased away from the true value (Skoog *et al.* 2017). To ensure accurate results, it is crucial to implement appropriate quality control measures, such as calibration procedures, standardization protocols, and proper instrument maintenance. Additionally, statistical techniques like error analysis, uncertainty quantification, and error propagation can help identify and mitigate the effects of both systematic and random errors (Ku 1966). The present study calculated the overall accuracy for each class by using field data given in table 4.

Table 3. Village-wise area for plantation cover in ha (500 m from the shoreline buffer).

Village name	Type of plantation						Grand total
	Cashew	Casuarina	Coconut	Mixed vegetation	Palm	Prosopis	
Ariankuppam	0.2	0.0	2.2	4.3	–	1.4	8.1
Kalapet	–	–	5.4	21.8	3.6	–	30.8
Kirumampakkam	–	8.1	4.2	5.6	1.1	2.6	21.5
Manapet	–	10.6	2.1	17.1	1.2	1.5	32.5
Manaveli	–	0.5	6.6	2.1	0.2	1.2	10.5
Pillaichavady	–	0.3	2.3	33.2	0.2	–	36.0
Pillayarkuppam	–	2.7	2.2	3.5	–	–	8.3
Pooranankuppam	3.3	2.1	10.0	15.1	1.9	0.9	33.3
Puducherry	–	–	2.0	22.9	0.2	0.1	25.1
Thengaithittu	–	2.0	–	–	–	11.5	13.4

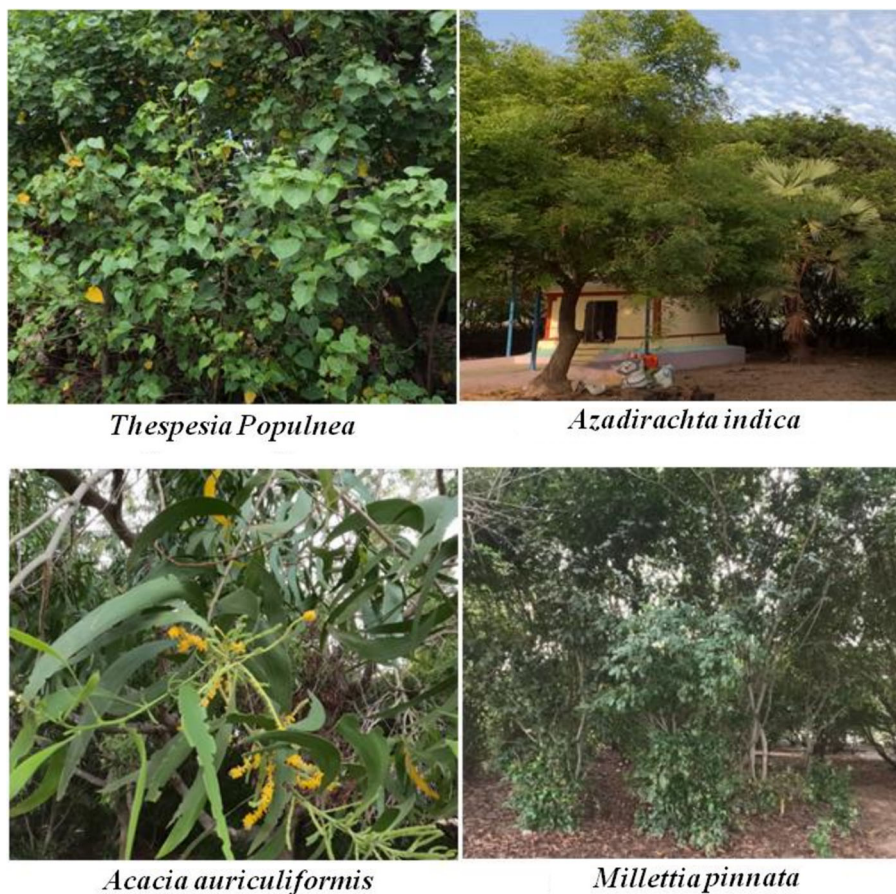


Figure 10. Dominant tree species found in the back dune, shoreline area of Pondicherry coast.

3.5 Threats

Prosopis and other invasive species are the natural threats to the bio shields in Pondicherry. Thengaithittu mangrove area is covered by Prosopis around 11.6 ha, as shown in figure 11. In addition, the sewage and household waste are mixed into the Thengaithittu water channel, which stops the free water flow along the necklace-shaped water channel

route. Anthropogenic activities such as coastal development, urbanization, tourism, and industrialization can degrade coastal habitats and disrupt plantation efforts. Pollution, habitat destruction, overharvesting of resources, and unregulated land use practices can contribute to the degradation of coastal ecosystems. Soil erosion can undermine the stability of coastal vegetation and inhibit plant growth by exposing roots and washing away

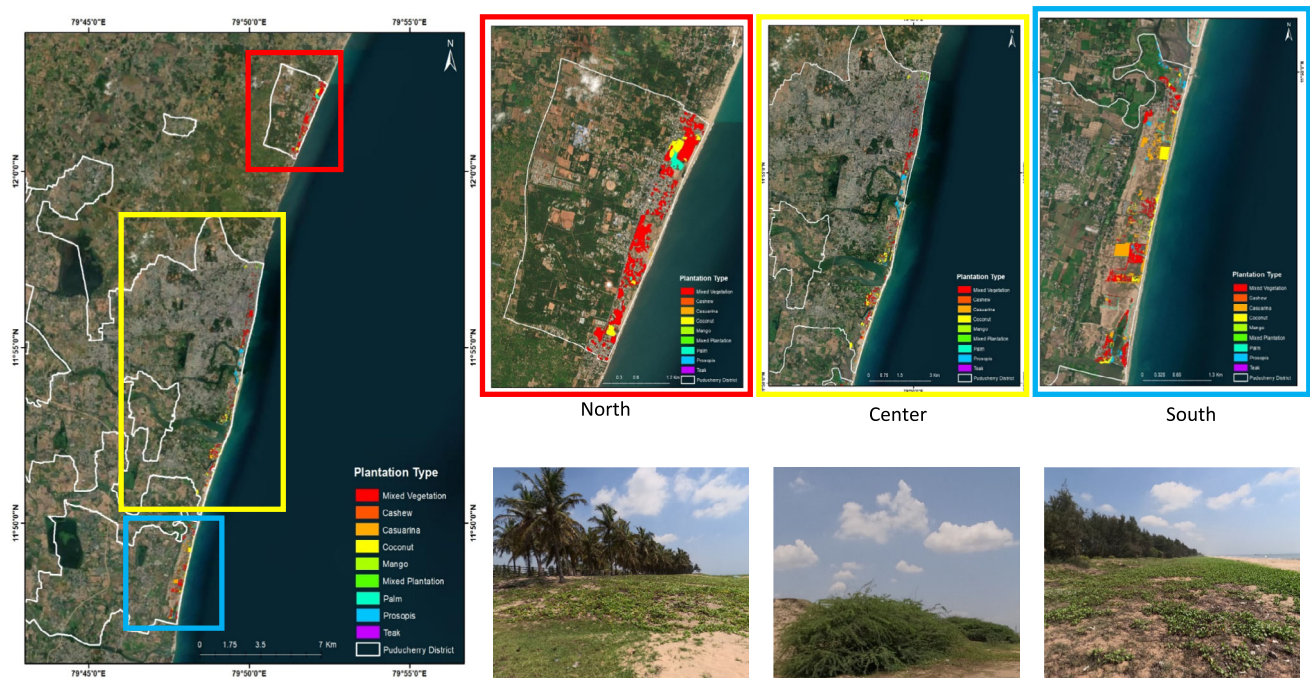


Figure 11. Spatial distribution of protective green belt plantation and dominant tree species found along the Pondicherry coast.

Table 4. Overall accuracy assessment for coastal green infrastructure mapping.

Class	Classified pixel	Reference site	Accuracy (%)
Plantation	27	30	90
Sand dune	20	20	100
Mangrove	26	30	86
Overall accuracy			92

nutrients. In addition, the eroding foreshore and damaged buildings due to manmade wave shifts were noticed in the southern part of Kalapet, adjacent to Tamil Nadu, as observed in figure 12(f, g). The natural dune locations on the Ariankuppam, Pondicherry and Oulgaret commune coasts were urbanized, and no dune exists currently.

Further, the existing beach was dredged for sewage disposal into the sea, which was recorded. The manmade dunes in Thengaithittu are also being removed and utilized. The anthropogenic impact of unorganized vehicle parking, boat parking and human pedestrian traffic causes severe vegetation fragmentation and damage in sandy backshore and sand dune vegetation (figures 4(b, c) and 10).

3.6 Discussion

Coastal zones are experiencing severe geomorphological changes due to natural and anthropogenic

influences on a global scale. Approximately 57% of the Pondicherry coast is under erosion, causing extensive human and economic loss through permanent inundation (Sundaravadivelu *et al.* 2005). Presently, this ecosystem is severely under major stress, such as mangrove habitat shift through invasive plant species, green shelterbelt fragmentation through various infrastructure developments, and natural and synthetic shoreline/sand dune breaches. The Pondy-necklace is one of the microhabitats with diverse mangroves that serve both ecosystem and tourism sector services. However, the ecosystem is under severe threat, such as the invasion of *Prosopis* sp. on mangrove wetlands. The mangrove ecosystem degradation through various anthropogenic activities is well documented (Ghosh *et al.* 2022; Akram *et al.* 2023). However, saline-tolerant intertidal ecosystem degradation through invasive *Prosopis* sp. has been noticed in recent decades. Plant invasion vulnerability in mangrove forests indicates the reduction in water and soil salinity through sewage input, disconnected seawater exchange by road network/encroachment and domesticated animals grazing (Harun-or-Rashid *et al.* 2009). Pondy-necklace backwater around Thengaithittu and Pooranakuppam was observed with water disconnection anthropogenic activities. *Prosopis* sp. was dominated in Arikkamedu.



Figure 12. (a) Habitat shift through invasive plant species. (b, c, d) Green shelterbelt fragmentation through boat parking, vehicle parking and infrastructure expansion. (e) Breach of shoreline and backshore/dune plantation through direct sewage flow and pedestrian traffic. (f, g) Erosion through shifting wave energy.

Pondicherry is a tourism-based city which received income of $\sim 2,16,495$ INR/year in 2023. The increased vehicle population and limited parking space resulted in an unorganized parking strategy and pedestrian traffic, impacting coastal shelter

plantation fragmentation. Coastal habitat fragmentation is one of the major factors contributing to biodiversity decline, and traffic is one of the principal concerns. The intense tourism activity concentration leads to coastal plantation

fragmentation/density thinning and sand dune loss (Bekker and Iuell 2003; Fantinato 2019). The pedestrian traffic mainly contributes to foredune degradation by destroying plant and community establishments (Sperandii *et al.* 2019). Rapid urbanization patterns in cyclone-prone coastal stretches of India are experiencing more disaster threats. Hence, 26.2% of the Pondicherry coast was managed through synthetic structures and ~24 km of coast facing low to medium erosion (Ramesh *et al.* 2011). The seawalls and protective groyne structures often cause negative impacts such as the ‘flanking effect and terminal groyne syndrome’ due to the dynamic and physical force of the waves, wind and tidal action of the coast (Zhou *et al.* 2023). The present study recorded the natural structures that safeguard the coastal area of Pondicherry.

4. Conclusion

The study demonstrates the efficacy of using ALTM data for mapping sand dunes along the Pondicherry coast. Detailed insights into dune morphology and distribution can inform coastal management strategies, including conservation efforts, shoreline stabilization measures, and land-use planning. Continued monitoring using LiDAR technology will be crucial for tracking coastal changes and implementing adaptive management practices. Combining GEE with ML techniques offers a robust and efficient approach to mapping coastal plantations. This methodology can be applied to other regions and land cover types, contributing to broader environmental monitoring and conservation efforts. As satellite imagery continues to improve in resolution and frequency, the accuracy and scalability of such mapping efforts are expected to enhance further, aiding in sustainable land management practices worldwide. Efforts to conserve and restore mangroves in Pondicherry are imperative for ensuring their long-term sustainability. This study presents a novel approach to identifying and mapping CGI using remote sensing data and machine learning techniques. While individual components like remote sensing, machine learning, and CGI mapping have been explored separately, the novelty of this study lies in the integration of these components into a comprehensive methodology tailored specifically for mapping CGI along the Pondicherry coast. This integrated approach and its application to a new

geographic area contribute to the advancement of knowledge in CGI mapping and coastal ecosystem management. Findings from this study will aid in strategizing policies and regulations to mitigate habitat destruction, promoting sustainable land use practices, and establishing protected areas and marine reserves. Community involvement and stakeholder engagement are crucial for successful conservation initiatives, fostering local stewardship and empowerment.

Furthermore, raising awareness about the importance of mangroves through education and outreach programs is essential for garnering public support and fostering a culture of conservation. There is a paucity of such studies along the Pondicherry coast. This study is unique in that it integrates different perspectives, such as coastal ecology, disaster management, coastal and environmental protection and policy.

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Author statement

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